Detector-device-independent quantum key distribution: From proof of principle to a high speed implementation

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QSIT Quantum Science and Technology
National Centre of Competence in Research

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FNSNF
Swiss National Science Foundation

Protected by quantum physics

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Practical QKD

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Practical QKD

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Quantum hacks

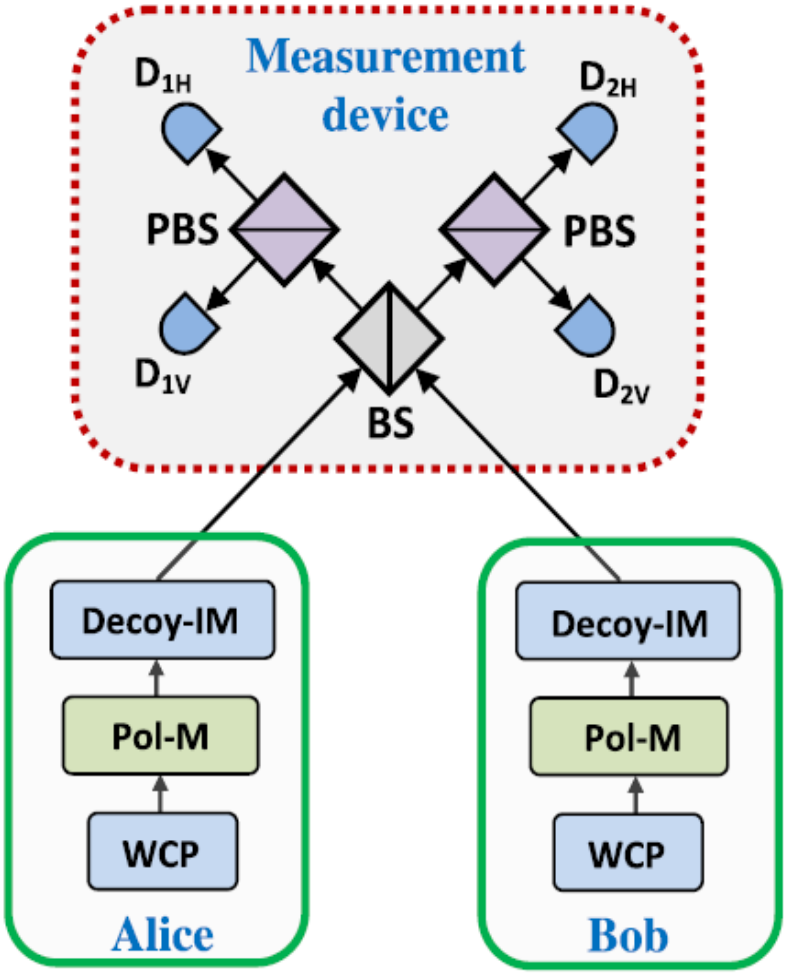
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Attack	Target component	Tested system
Time-shift	Detector	Commercial
Time-information	Detector	Research
Detector control	Detector	Commercial, research
Detector dead time	Detector	Research
Channel calibration	Detector	Commercial
Phase remapping	Phase modulator	Commercial
Faraday Mirror	Faraday mirror	Theory
Wavelength	Beamsplitter	Theory
Phase information	Source	Research
Device calibration	Local oscillator	Research



- H.-K. Lo, M. Curty, K. Tamaki "*Secure quantum key distribution*," Nature Photonics 8, 595604 (2014)

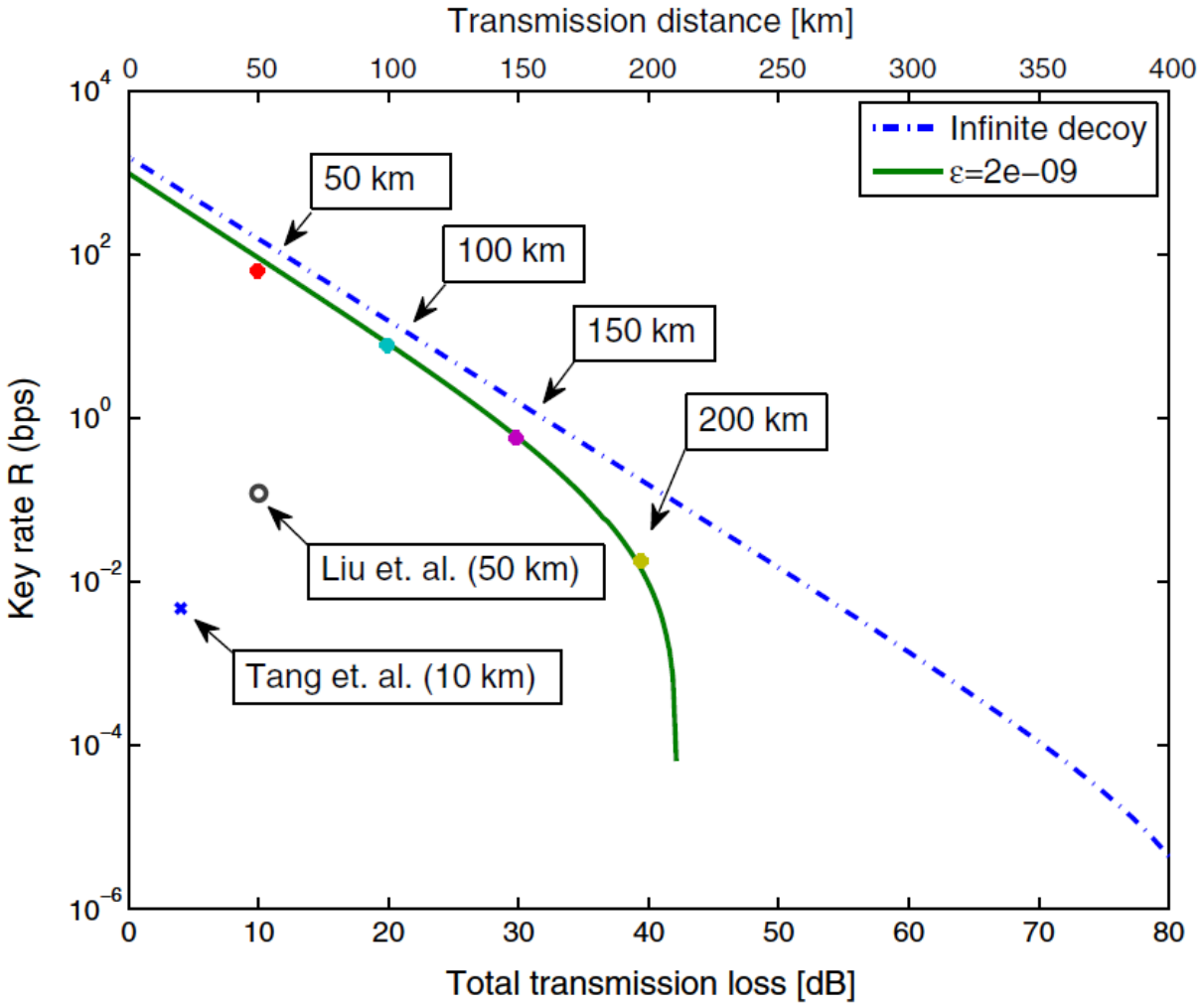
MDI-QKD



- BSM
  - 50% efficient at best
- 2 photons
  - Low rates
  - Have to be indistinguishable
- Finite key yield not as high as prepare and measure protocols

- H.-K. Lo et al., "*Measurement-device-independent quantum key distribution*," PRL 108, 130503 (2012)

Performance



- MDI
  - 40% detection efficiency
  - 50 km: 67 bps
  - **200 km: 0.018 bps**
    - 130 hours
- P&M QKD
  - Coherent one-way protocol
  - **200 km: 1020 bps**
  - 307 km: 3.2 bps
    - 4.5 hours

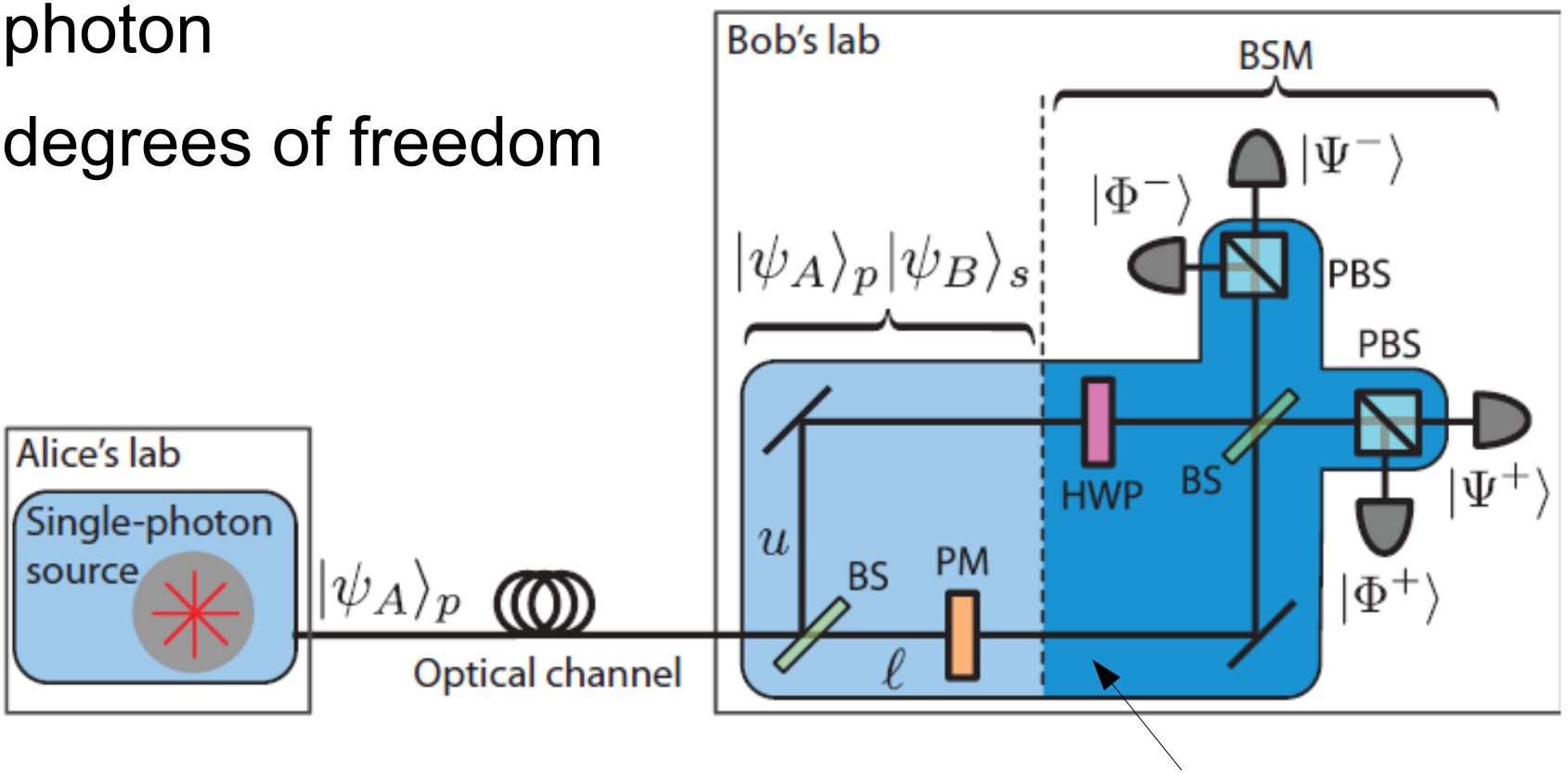
- Y.-L. Tang et al., "*Measurement device independent quantum key distribution over 200 km*," PRL 113, 190501 (2014)
- B. Korzh, C. W. Lim et al., "*Provably Secure and Practical Quantum Key Distribution over 307 km of Optical Fibre*," Nature Photonics 9, 163-168 (2015)

DDI-QKD

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Detector-device-independent

- 1 photon
- 2 degrees of freedom



- C.C.W. Lim, B. Korzh et al., "*Detector-device-independent quantum key distribution*" APL 105, 221112 (2014)

Protocol – raw key

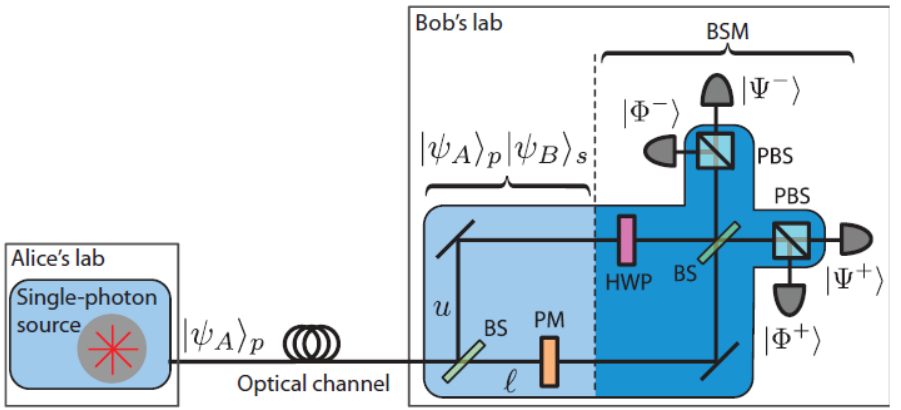
1. Alice chooses state
   - encodes bit value
2. Bob chooses state
3. Bob announces a BSM
4. Bob reveals basis
5. Alice confirms if compatible (sifting)
6. Bob determines Alice's bit value

$$|\psi_A\rangle_p \in_r \begin{cases} |+\rangle &= \frac{1}{\sqrt{2}}(|H\rangle + |V\rangle), \\ |-\rangle &= \frac{1}{\sqrt{2}}(|H\rangle - |V\rangle), \\ |+i\rangle &= \frac{1}{\sqrt{2}}(|H\rangle + i|V\rangle), \\ |-i\rangle &= \frac{1}{\sqrt{2}}(|H\rangle - i|V\rangle), \end{cases}$$

$$|\psi_B\rangle_s = (|u\rangle + \mathrm{e}^{i\varphi}|\ell\rangle)$$

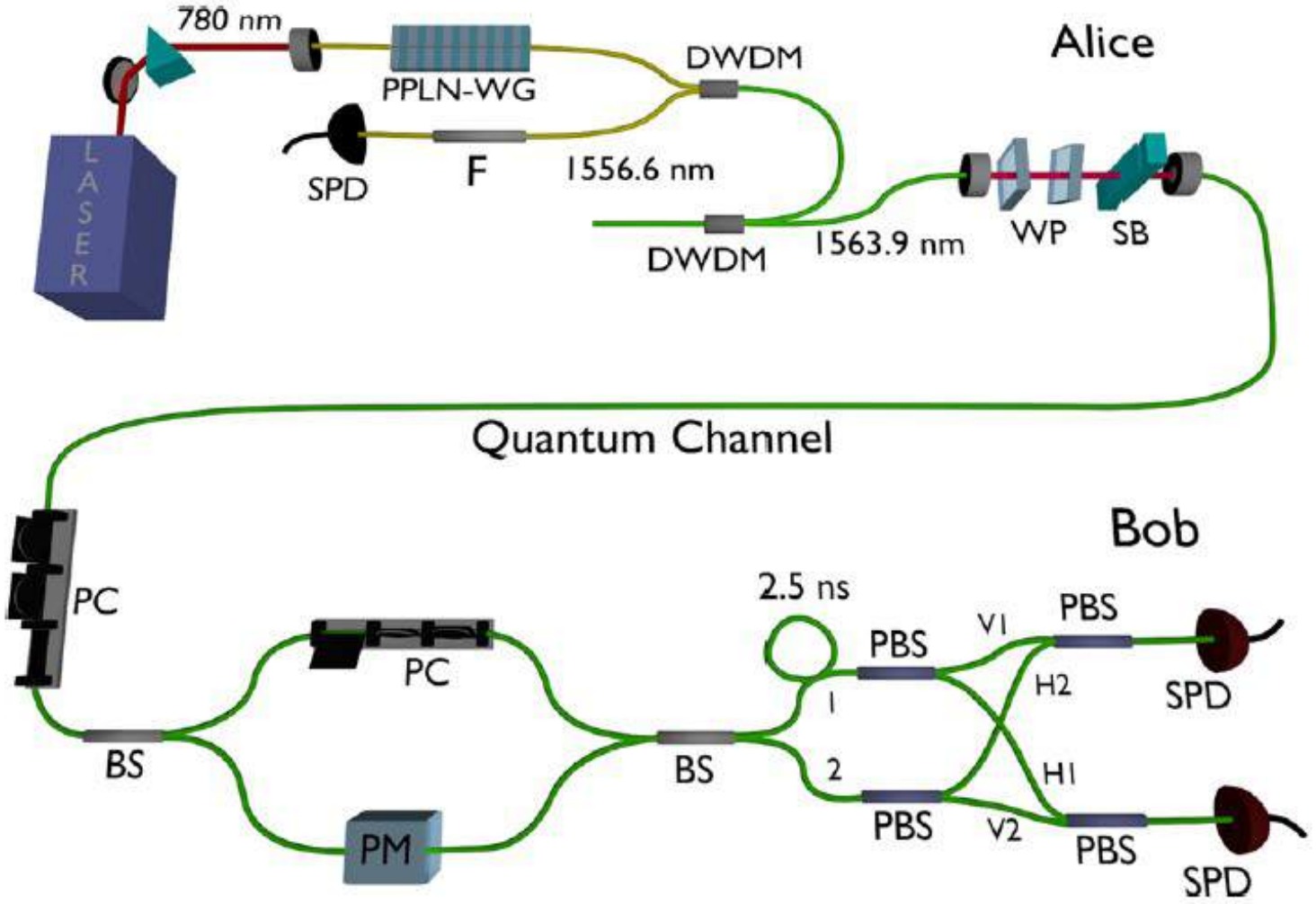
$$|\Phi^{\pm}\rangle = \frac{1}{\sqrt{2}}(|H\rangle_p|u\rangle_s \pm |V\rangle_p|\ell\rangle_s)$$

$$|\Psi^{\pm}\rangle = \frac{1}{\sqrt{2}}(|H\rangle_p|\ell\rangle_s \pm |V\rangle_p|u\rangle_s)$$



- C.C.W. Lim, B. Korzh et al., "*Detector-device-independent quantum key distribution*" APL 105, 221112 (2014)

Proof of principle



- C.C.W. Lim, B. Korzh et al., "*Detector-device-independent quantum key distribution*" APL 105, 221112 (2014)

Results

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(a) $|\Phi^+\rangle$

	$+$	$-$	$+i$	$-i$
$+$	0.49	0.01	0.25	0.26
$-$	0.01	0.50	0.25	0.27
$+i$	0.27	0.26	0.01	0.48
$-i$	0.24	0.23	0.50	0.01

(b) $|\Psi^+\rangle$

	$+$	$-$	$+i$	$-i$
$+$	0.49	0.02	0.25	0.27
$-$	0.00	0.50	0.27	0.24
$+i$	0.29	0.23	0.49	0.00
$-i$	0.23	0.25	0.01	0.55

(c) $|\Psi^-\rangle$

	$+$	$-$	$+i$	$-i$
$+$	0.00	0.48	0.28	0.25
$-$	0.54	0.00	0.25	0.23
$+i$	0.25	0.26	0.01	0.52
$-i$	0.26	0.24	0.50	0.01

(d) $|\Phi^-\rangle$

	$+$	$-$	$+i$	$-i$
$+$	0.00	0.47	0.25	0.25
$-$	0.54	0.00	0.23	0.25
$+i$	0.26	0.26	0.48	0.00
$-i$	0.26	0.21	0.00	0.56

Low QBER

- 1.5%

Full BSM

Coincidence rate

- 50 cps



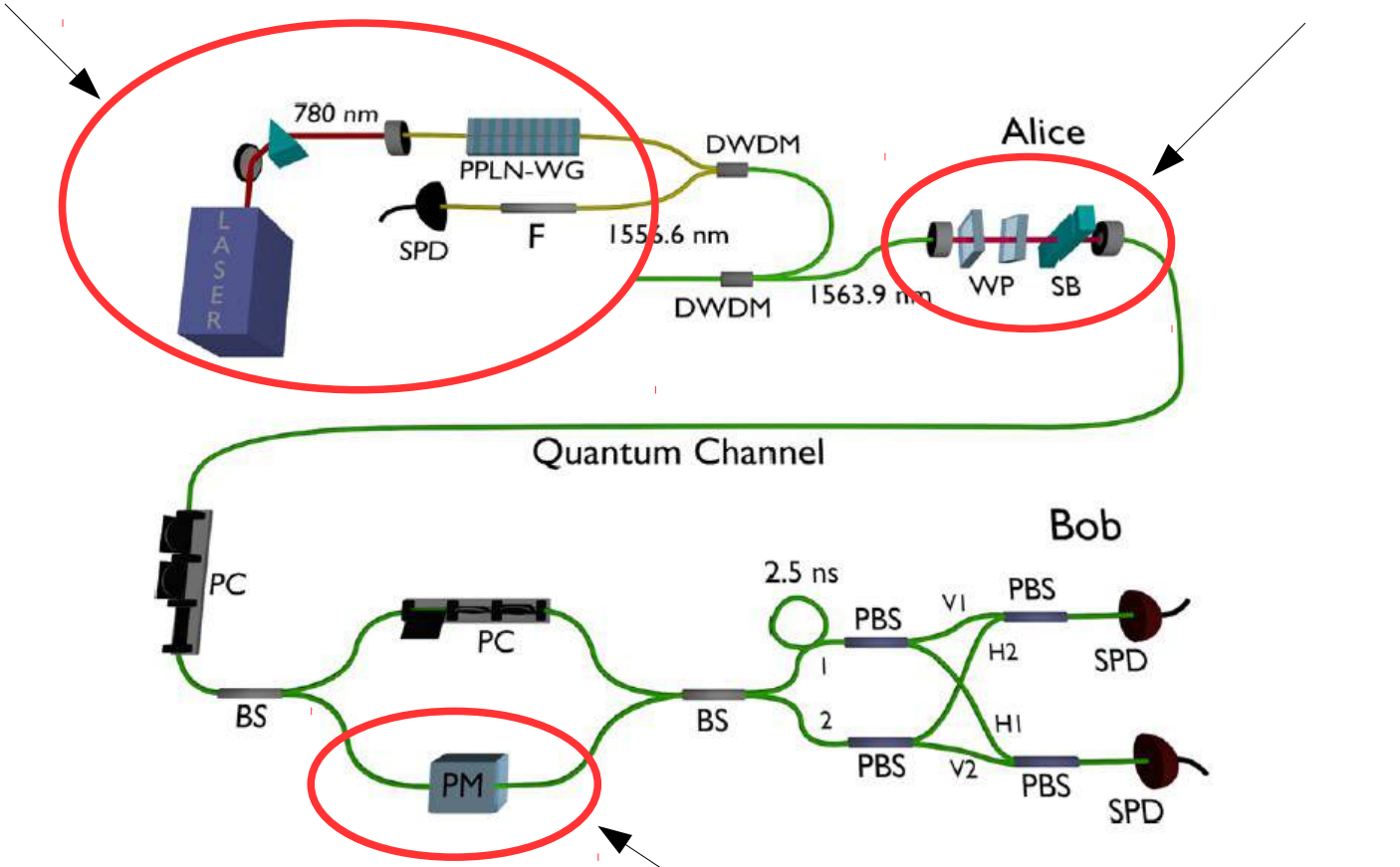
- C.C.W. Lim, B. Korzh et al., "*Detector-device-independent quantum key distribution*" APL 105, 221112 (2014)

Real system wish list

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- Weak coherent pulses
- Decoy state source

- Real time modulation

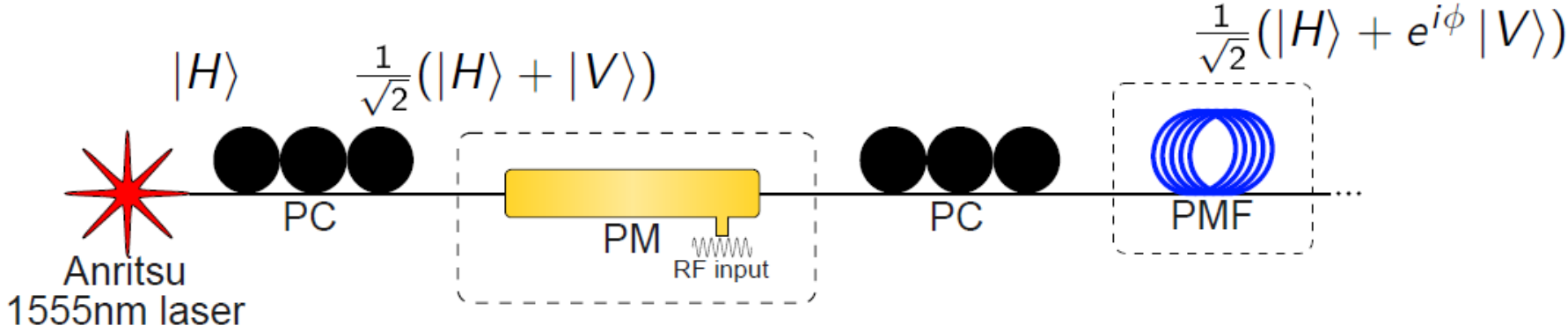


- High speed modulation
- Polarization insensitive

Alice

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Fast polarization modulation

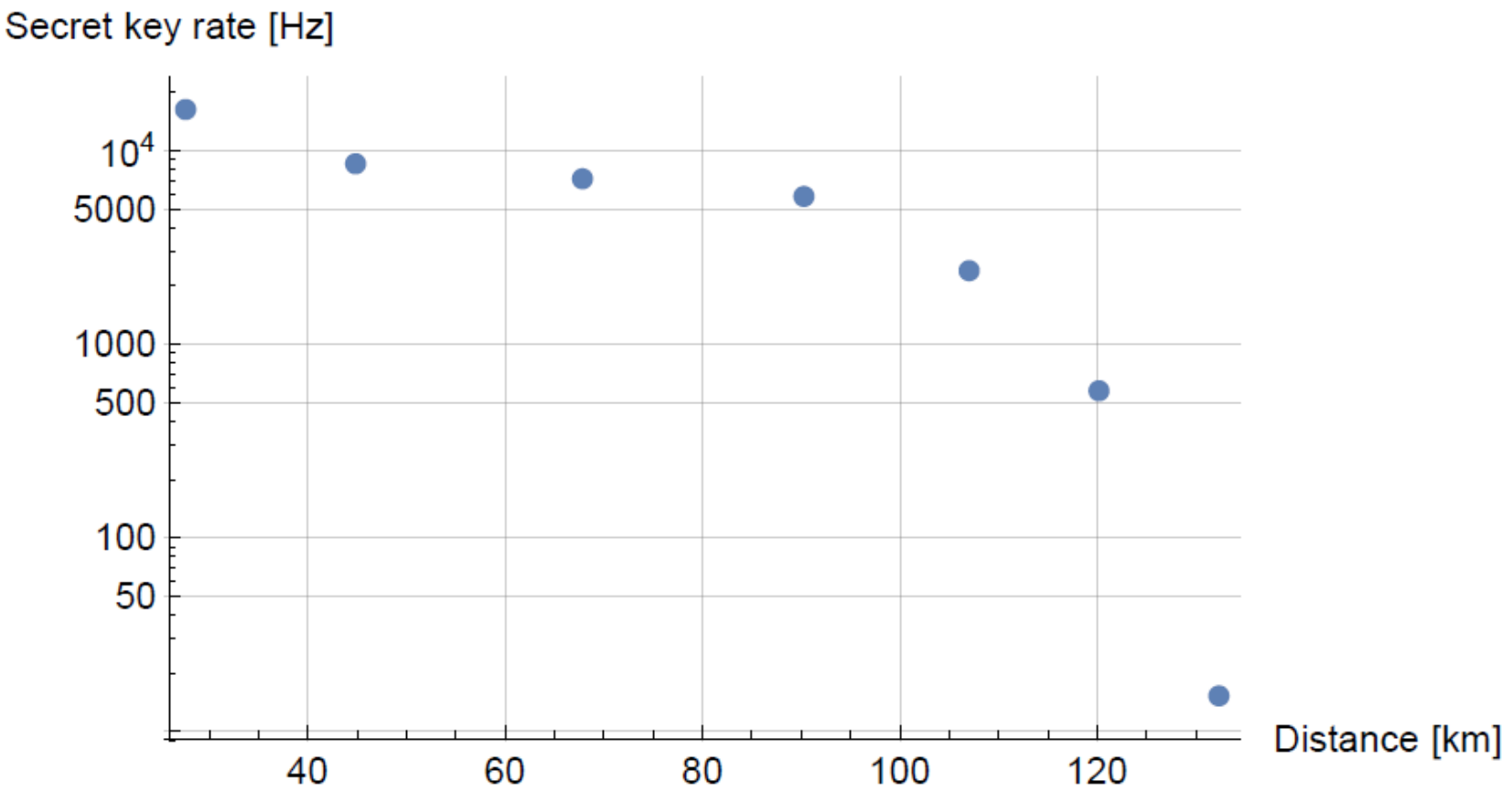


- 625 MHz
- Real time operation
  - Encoding, sifting distillation
  - 2.5 GHz FPGA platform
- Phase randomized weak coherent pulses
- Decoy encoding can be added with intensity modulator



- N. Walenta et al, "*A fast and versatile quantum key distribution system with hardware key distillation and wavelength multiplexing*," New J. Phys. 16, 013047 (2014)
- B. Korzh, C. W. Lim et al., "*Provably Secure and Practical Quantum Key Distribution over 307 km of Optical Fibre*," Nature Photonics 9, 163-168 (2015)

BB84 test



Optimum QBER

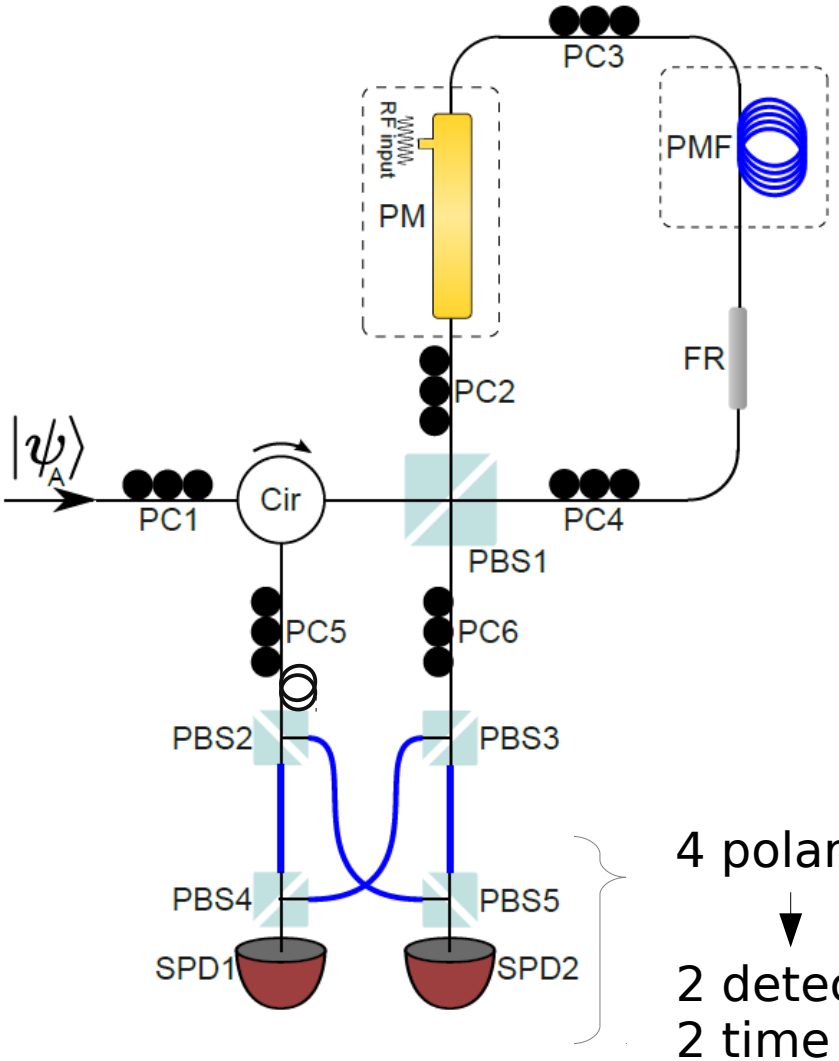
Q_z= 1.2%

Q_x= 6.4%

Weak coherent states

- no decoys

DDI - Bob



Stirling cooled InGaAs/InP Detectors

- DCR = 1 - 50 cpc @ 20%
- 2 detectors in 1 cooler



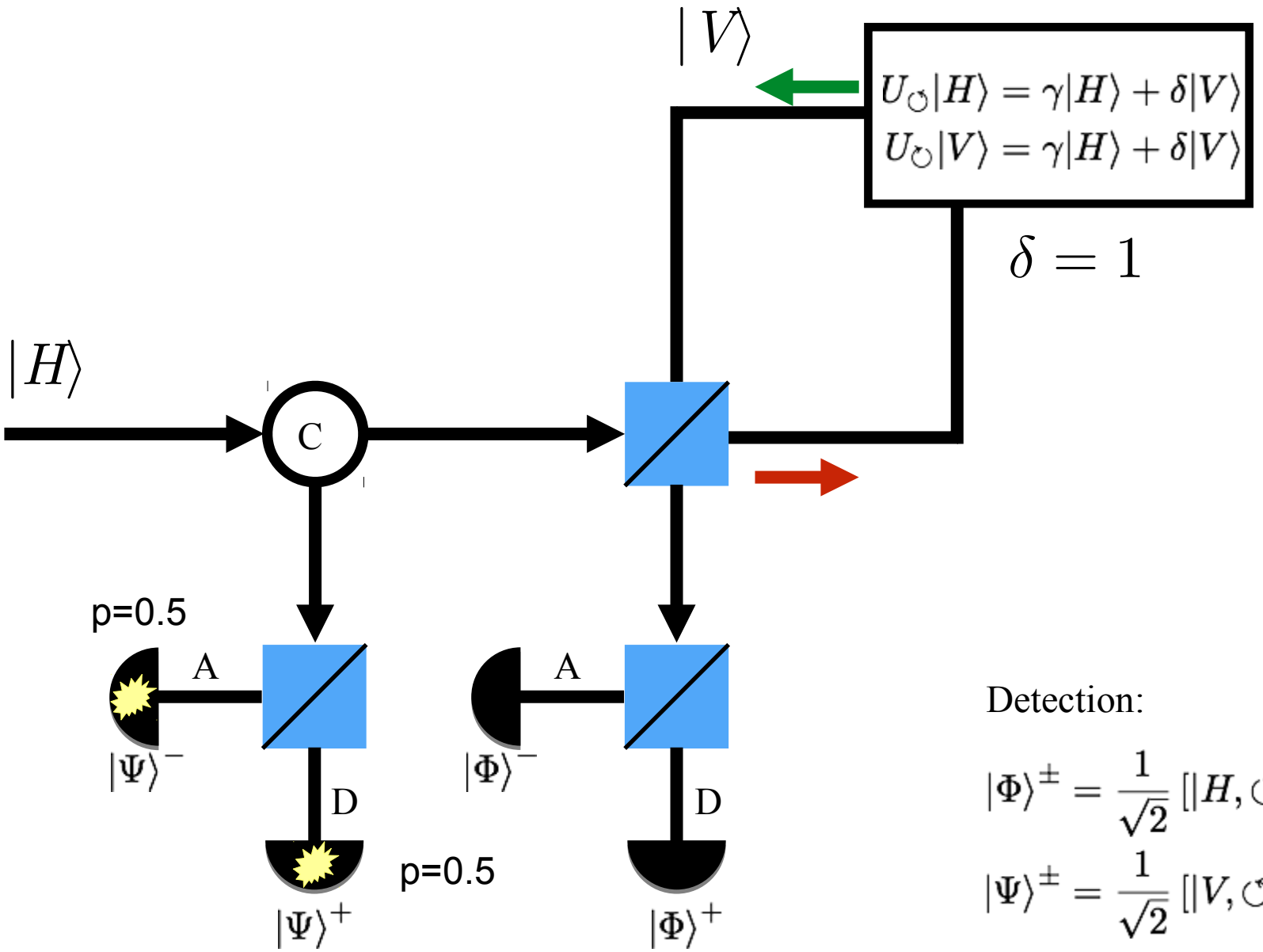
4 polarization ports

↓

2 detectors
2 time bins (800 ps separation)

- B. Korzh et al., *"Free-running InGaAs single photon detector with 1 dark count per second at 10% efficiency,"* Appl. Phys. Lett. 104, 081108 (2014)

DDI - Bob

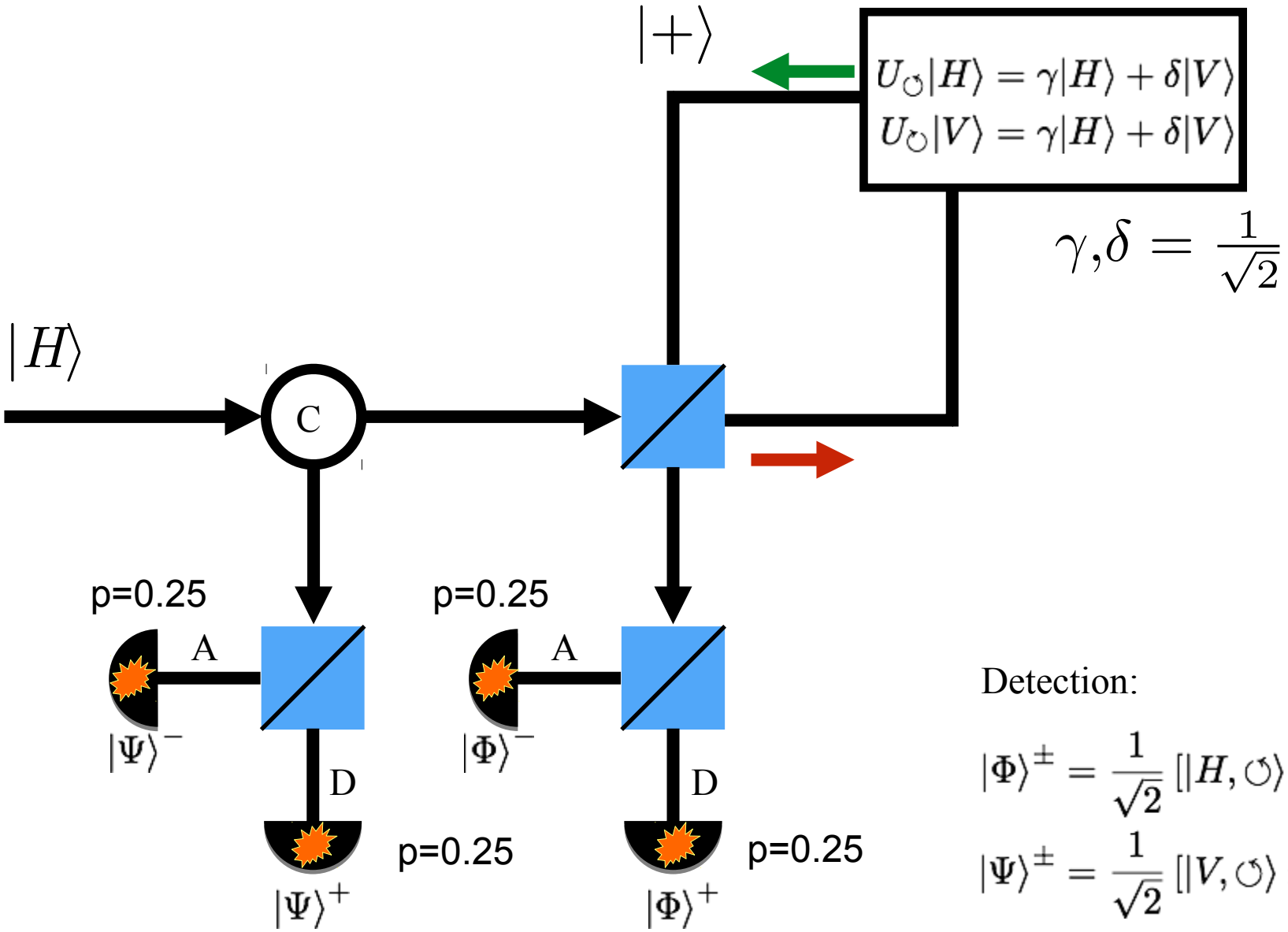


Detection:

$$|\Phi\rangle^{\pm} = \frac{1}{\sqrt{2}}[|H,\circlearrowleft\rangle \pm |V,\circlearrowright\rangle]$$

$$|\Psi\rangle^{\pm} = \frac{1}{\sqrt{2}}[|V,\circlearrowleft\rangle \pm |H,\circlearrowright\rangle]$$

DDI - Bob

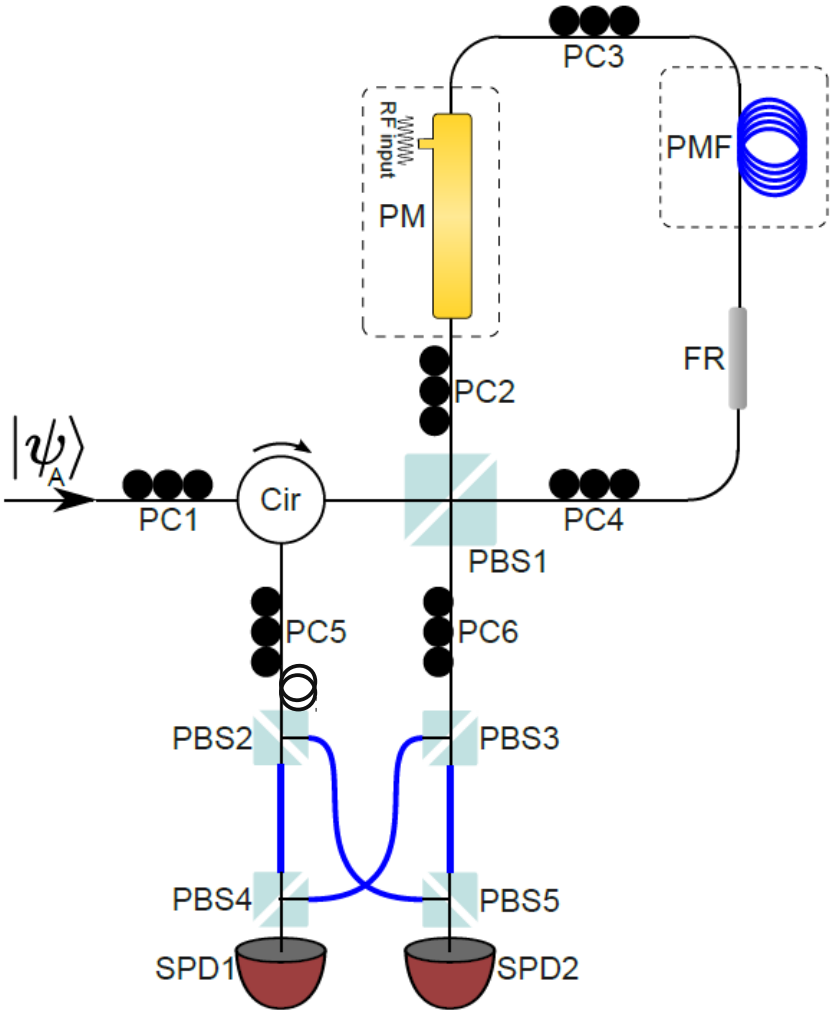


Detection:

$$|\Phi\rangle^{\pm} = \frac{1}{\sqrt{2}}[|H,\circlearrowleft\rangle \pm |V,\circlearrowright\rangle]$$

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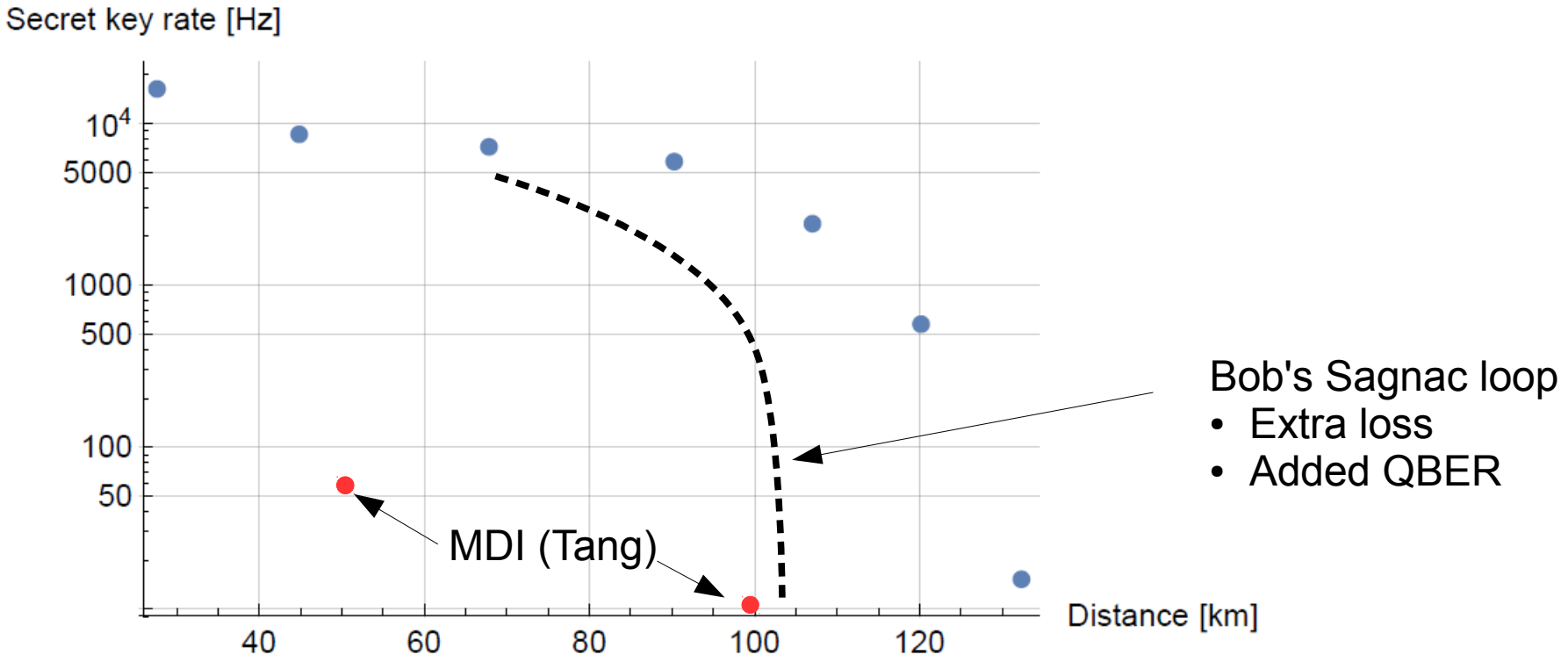
DDI - Bob



Measured visibility:
92 %

Using fixed sequences
at 625 MHz

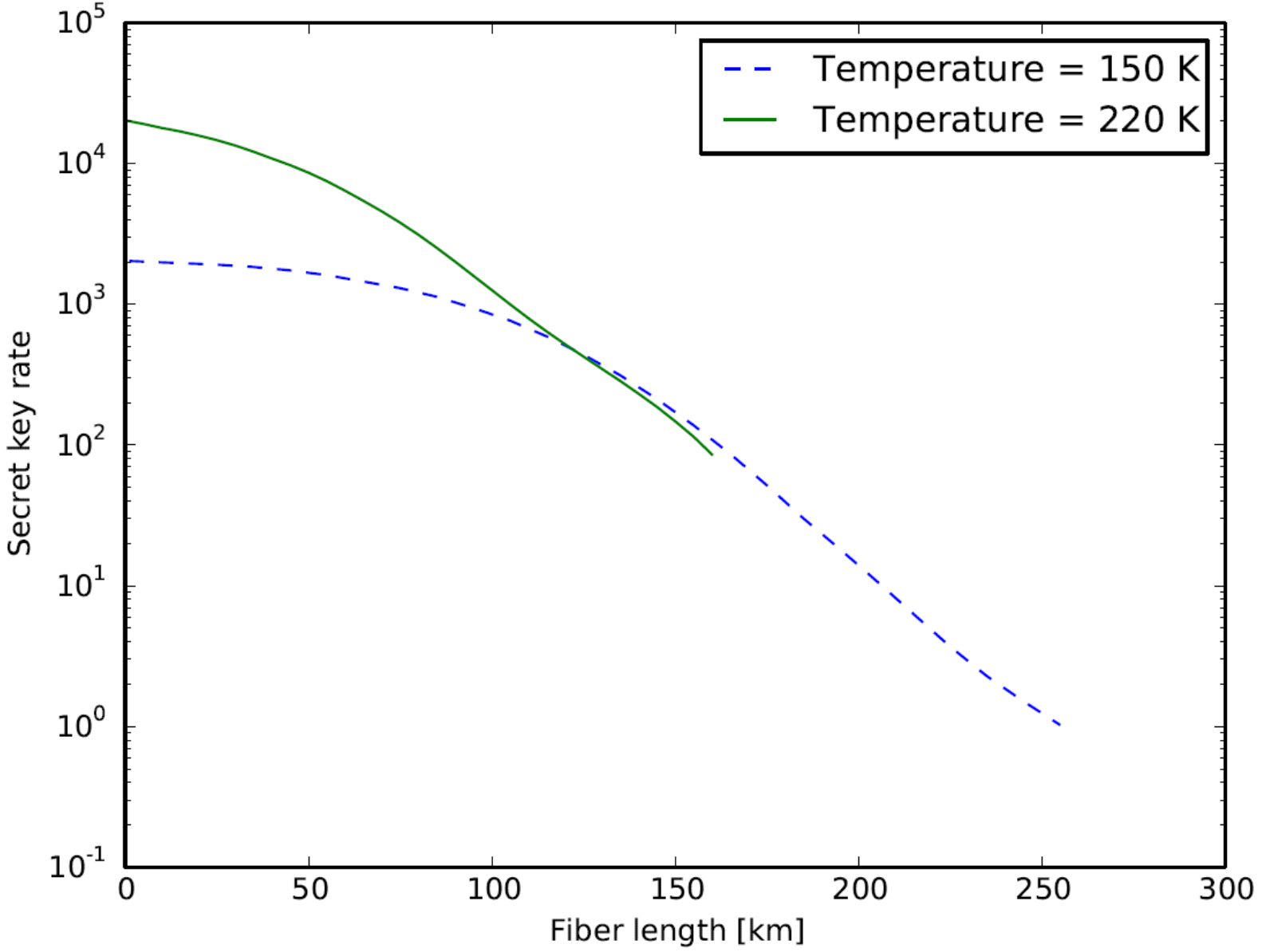
Expected performance



Weak coherent states
- no decoys

Decoy state DDI-QKD

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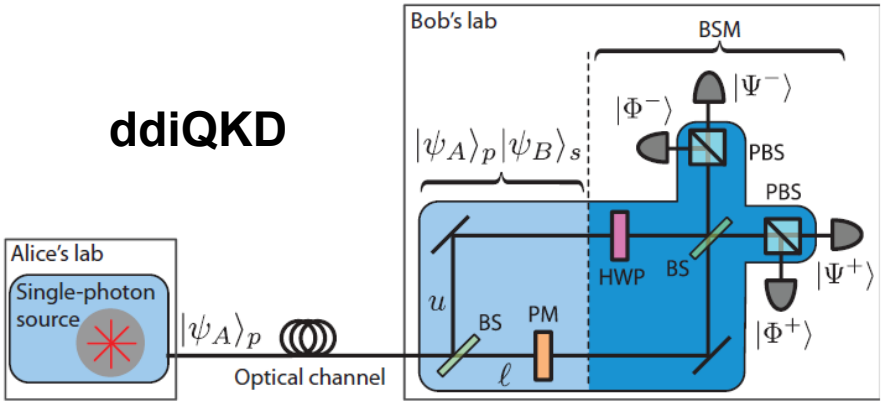


Conclusions

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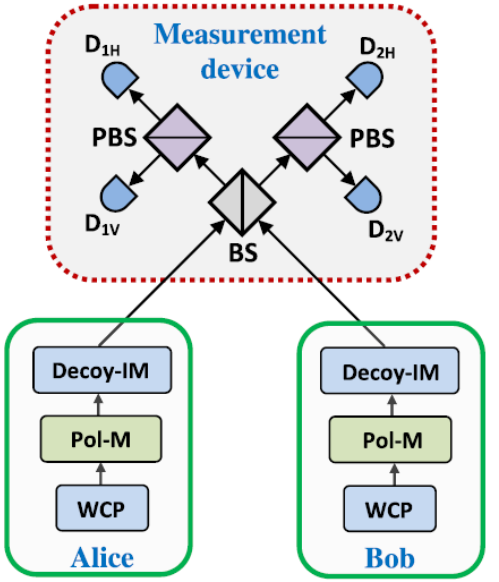
- DDI is a simpler alternative to MDI
  - Higher performance
  - More assumptions
    - Trusted linear optics
  - Only the weakest device left characterized
    - Detectors

ddiQKD



- Demonstration of proof of principle
- High speed, real-time system developed

mdiQKD



- Other solutions to detector hacking might prove practical also

- B. Qi, "*Trustworthiness of detector in quantum key distribution with untrusted detector*," PRA 91, 020303(R) (2015)
- P. Gonzalez et al., "*Quantum key distribution with untrusted detectors,*" PRA 92, 022337 (2015)

Thank you



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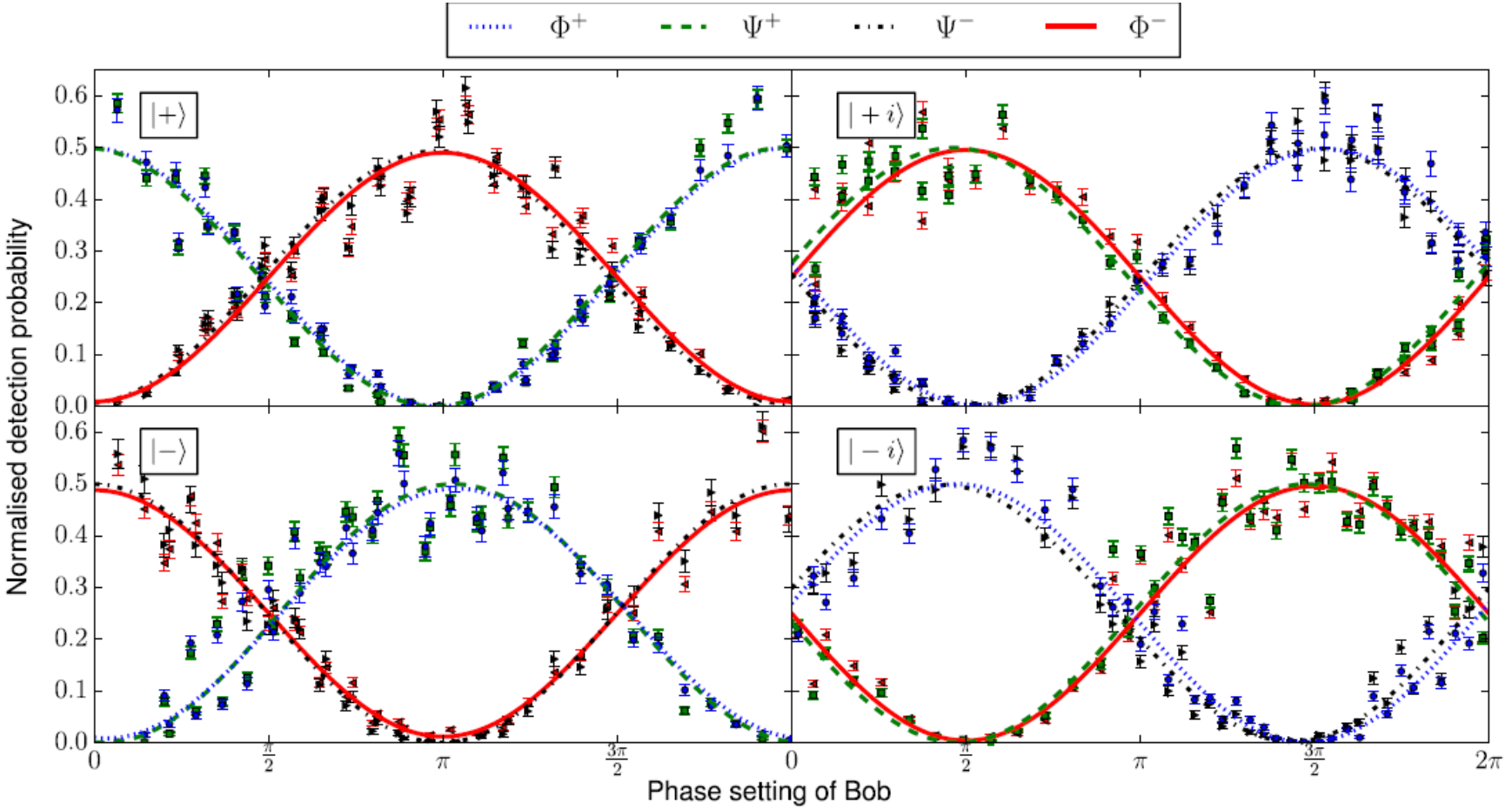




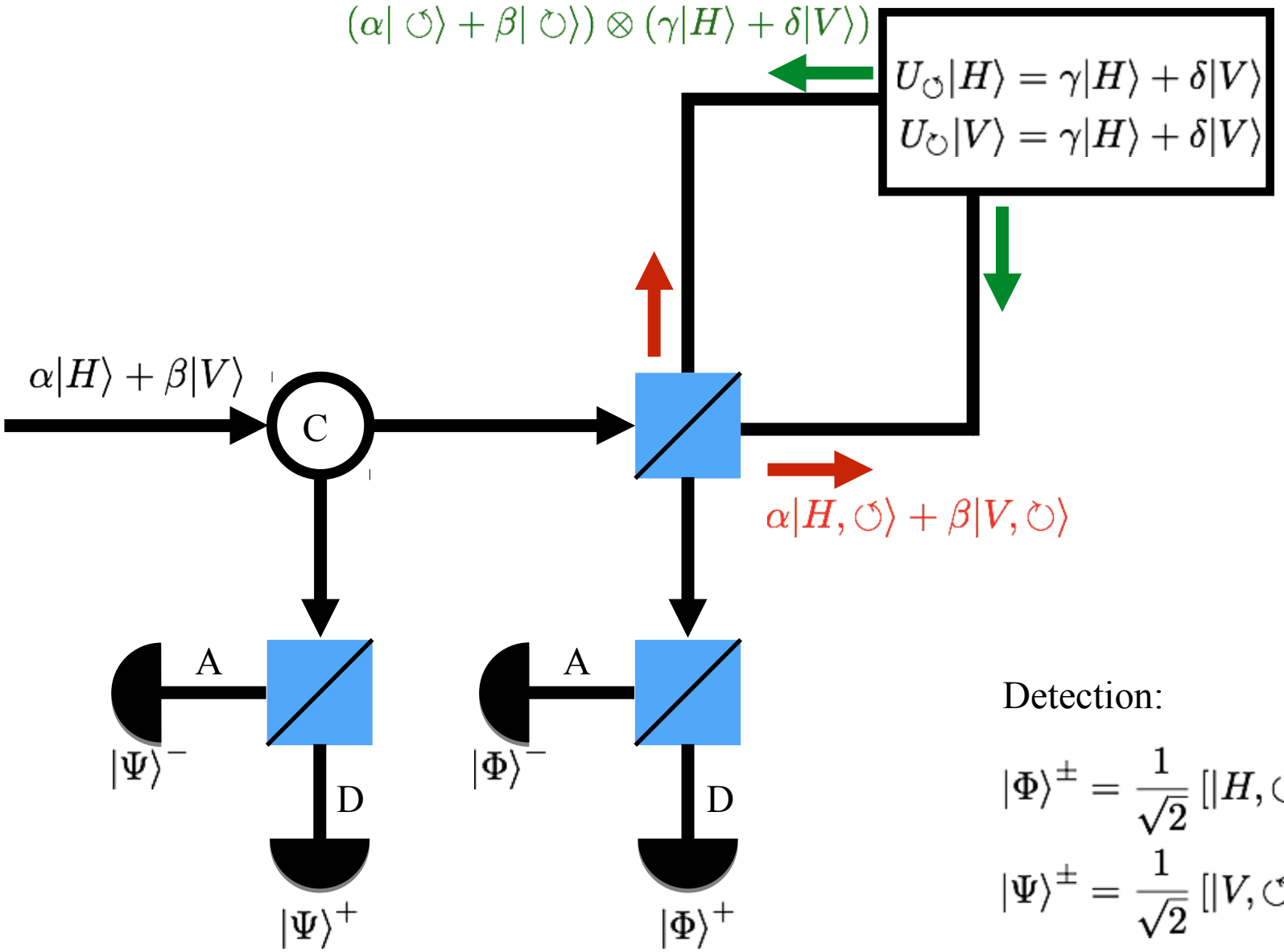
[1] Group of Applied Physics, University of Geneva, Switzerland
[2] Oak Ridge National Laboratory, Oak Ridge, TN, USA

Proof of principle

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DDI - Bob

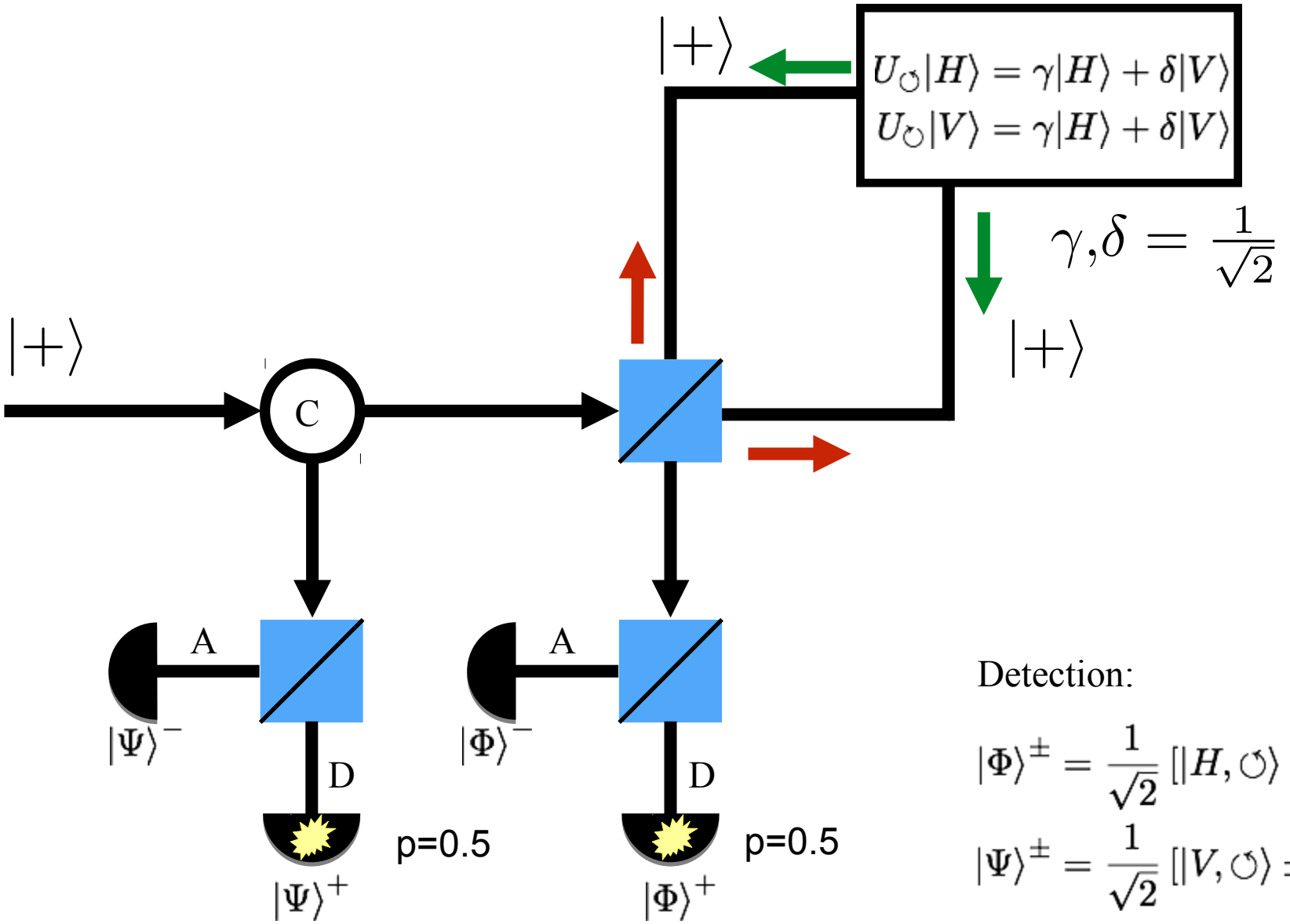


Detection:

$$|\Phi\rangle^{\pm} = \frac{1}{\sqrt{2}}\left[|H,\circlearrowleft\rangle \pm |V,\circlearrowright\rangle\right]$$

$$|\Psi\rangle^{\pm} = \frac{1}{\sqrt{2}}\left[|V,\circlearrowleft\rangle \pm |H,\circlearrowright\rangle\right]$$

DDI - Bob



Detection:

$$|\Phi\rangle^{\pm} = \frac{1}{\sqrt{2}}\left[|H,\circlearrowleft\rangle \pm |V,\circlearrowright\rangle\right]$$

$$|\Psi\rangle^{\pm} = \frac{1}{\sqrt{2}}\left[|V,\circlearrowleft\rangle \pm |H,\circlearrowright\rangle\right]$$

Notable QKD demonstrations

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	Quantum channel		Detector			Security			
	Distance (km)	Attenuation (dB)	Type	Temperature (K)	Protocol	Attacks	Finite-key size	ϵ_{qkd}	r_{sec} (bps)
This work	307	51.9	InGaAs‡	153	COW	Collective	6.6×10^5	4×10^{-9}	3.18
Wang 2012[24]	260	52.9	SNSPD	1.7	DPS	Individual	-	-	1.85
Stucki 2009[25]	250	42.9	SNSPD	2.5	COW	Collective	-	-	15
Takesue 2007[26]	200	42.1	SNSPD	3	DPS	Individual	-	-	12.1
Liu 2010[27]	200	-	SNSPD	2.4	BB84	Collective	-	-	15
Rosenberg 2009[28]	135	27.8	SNSPD	3	BB84	Collective	-	-	0.2
Namekata 2011[29]	160	33.6	InGaAs§	193	DPS	Individual	-	-	490
Yuan 2009[30]	100	20	InGaAs§	243	BB84	Collective	-	-	1.01×10^4
Shimizu 2014[23]	90*	30	SNSPD	2.5	DPS	Individual	-	-	1,100
Lucamarini 2013[19]	80†	16	InGaAs§	243	BB84	Collective	10^9	10^{-10}	1.20×10^5
Walenta 2014[18]	25†	5.3	InGaAs§	293	COW	Collective	10^6	4×10^{-9}	2.25×10^4

Finite key effects ignored:

- Misleading maximum distance

A quantifiable security statement is crucial for QKD

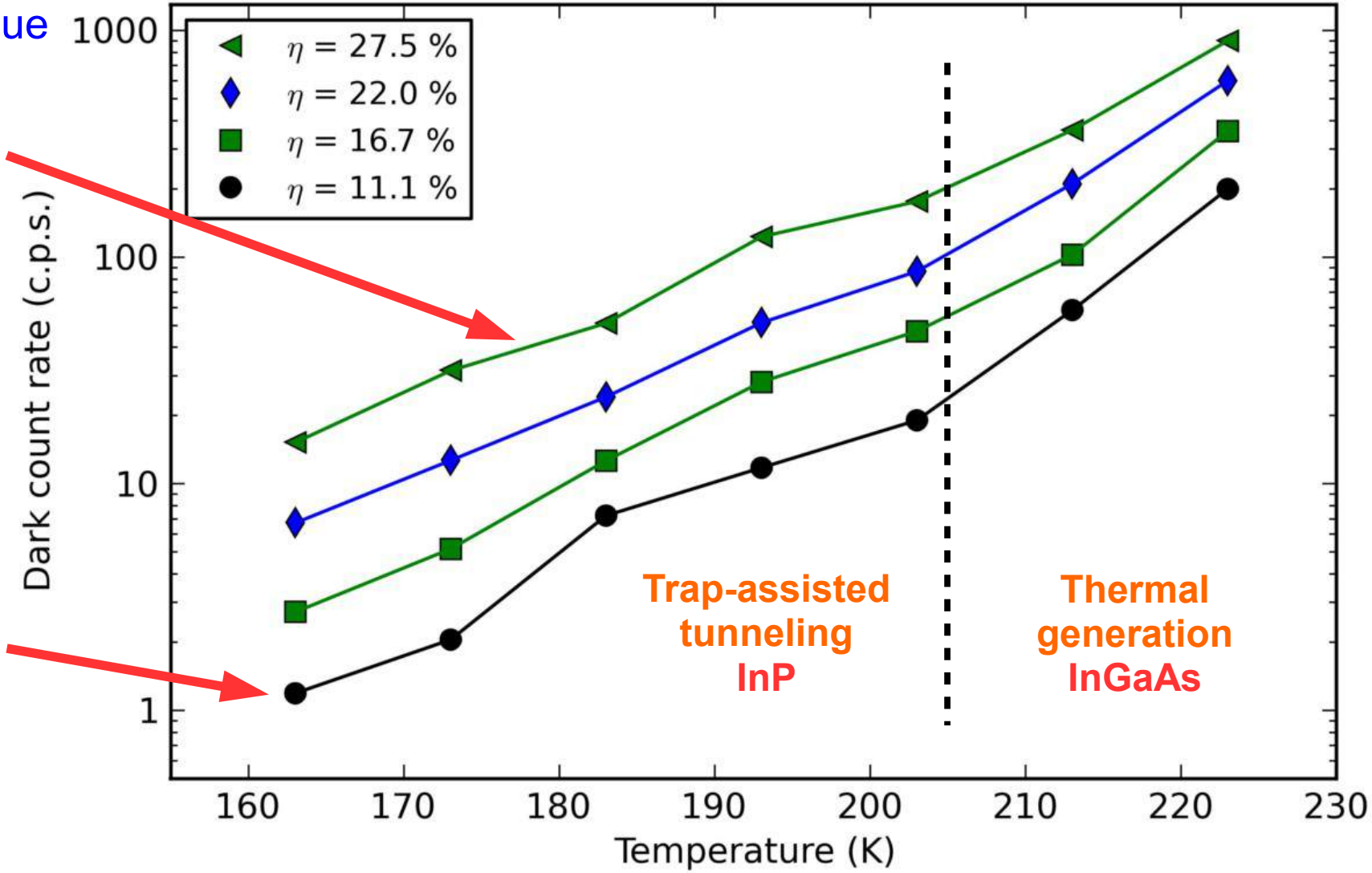
- B. Korzh, C. W. Lim et al., "*Provably Secure and Practical Quantum Key Distribution over 307 km of Optical Fibre*," Nature Photonics 9, 163-168 (2015)

Dark count rate

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Reduction due to lower breakdown voltage => smaller field

1.2 cps reduction of 2 o.m.

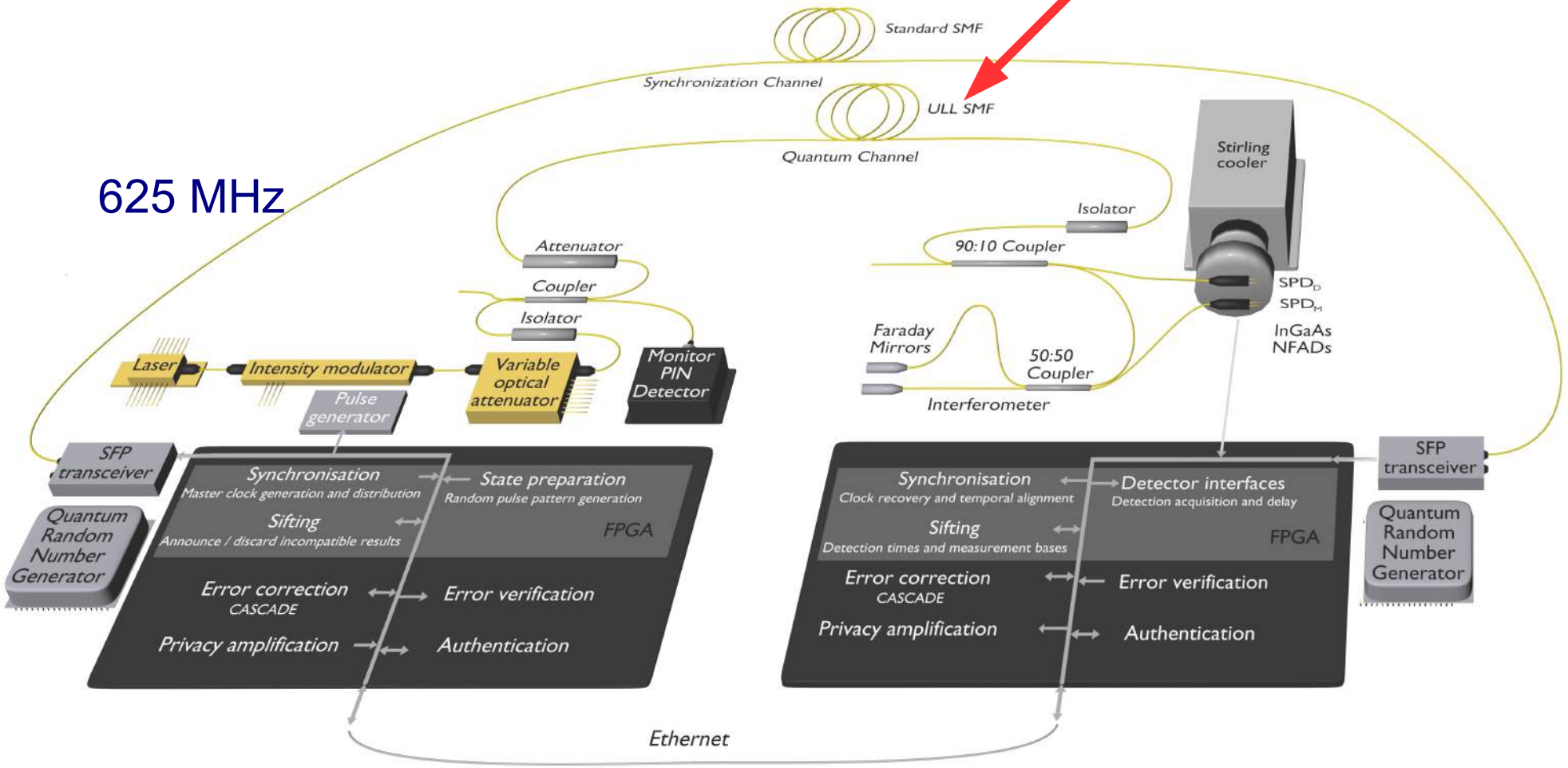


- B. Korzh et al., *"Free-running InGaAs single photon detector with 1 dark count per second at 10% efficiency,"* Appl. Phys. Lett. 104, 081108 (2014)

Experimental setup

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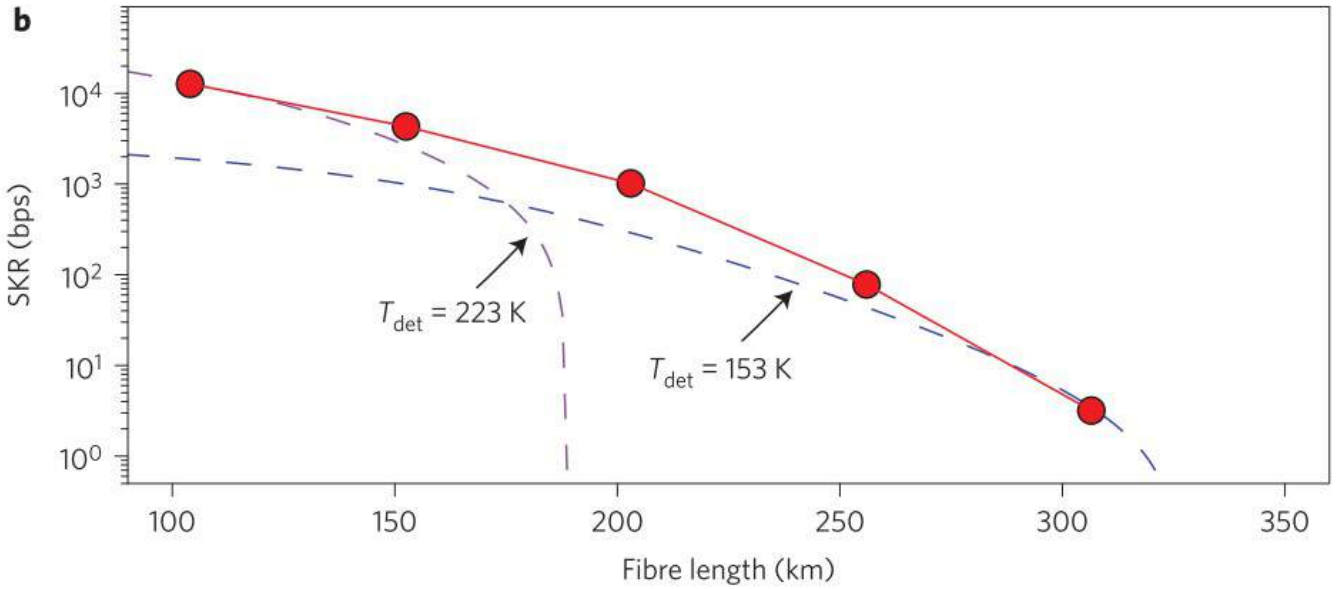
Ultra low loss Corning fiber: 0.16 dB/km

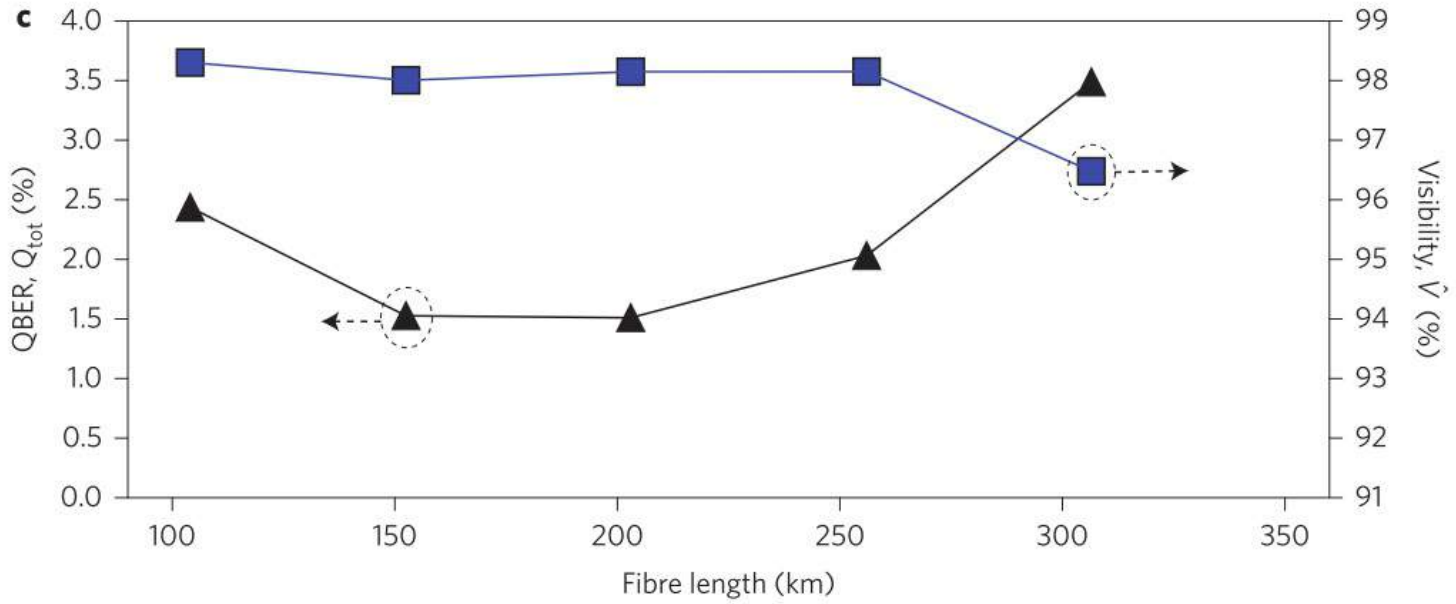


- B. Korzh, C. W. Lim et al., "*Provably Secure and Practical Quantum Key Distribution over 307 km of Optical Fibre*," Nature Photonics 9, 163-168 (2015)

Secret key rates

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VIEW

The Physics arXiv Blog
May 17, 2010

Commercial Quantum Cryptography System Hacked

Physicists have mounted the first successful attack of its kind on a commercial quantum cryptography system.

ADAM MANN SCIENCE 06.07.13 6:30 AM

LAWS OF PHYSICS SAY QUANTUM CRYPTOGRAPHY IS UNHACKABLE. IT'S NOT